

Introduction To Phylogeny

Madeleine DE SOUSA VIOLANTE





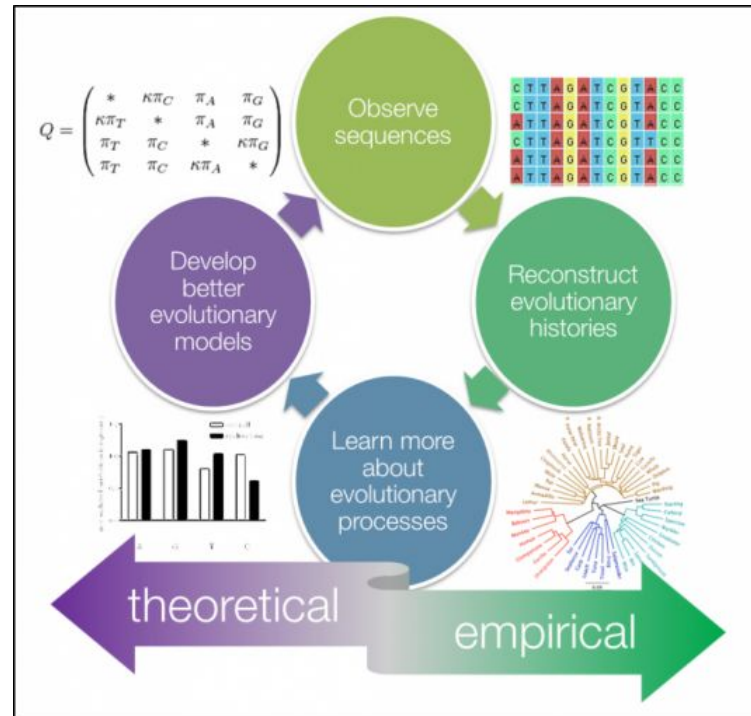
Learning outcomes

- Identify the various sections of a phylogenetic tree
- Define various phylogenetic terminology
- Describe the process of parsimony tree building
- Translate a distance matrix into a phylogeny
- Describe the process of tree space searching
- Recognise the maximum likelihood, bayesian & bootstrap approaches

What is phylogenetics

Phylogenetics is the study of evolutionary relationships among biological entities – often species, individuals or genes (which may be referred to as taxa).

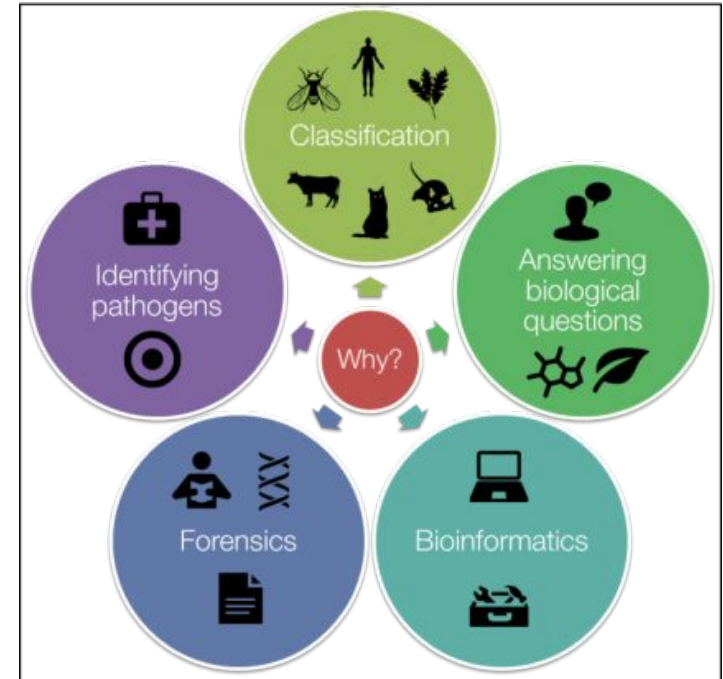
- What are the evolutionary relationships or histories among my species/individuals/genes of interest?
- How do sequences evolve?
- Can I better describe processes of sequence evolution with a mathematical model?



Why is phylogenetics important?

It enriches our understanding of how genes, genomes, species evolve.

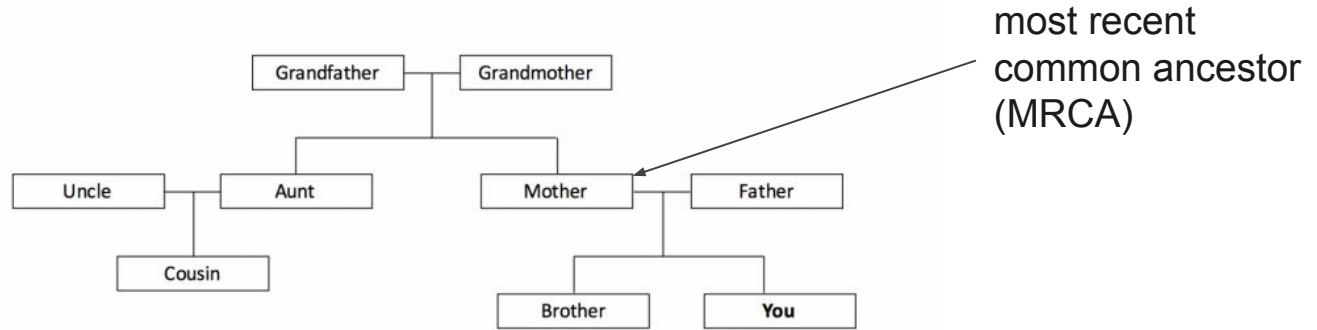
- more accurate descriptions of patterns of relatedness
- assess DNA evidence (crime, food contamination)
- inform conservation policy
- finding out about which species the pathogen is related to and subsequently the likely source of transmission



Charles Darwin



Example of a family tree



“True tree” = complete explanation of the genealogical relationships

- all of the ancestors and descendants who are or were recently living
- the identities of the biological parents of each child

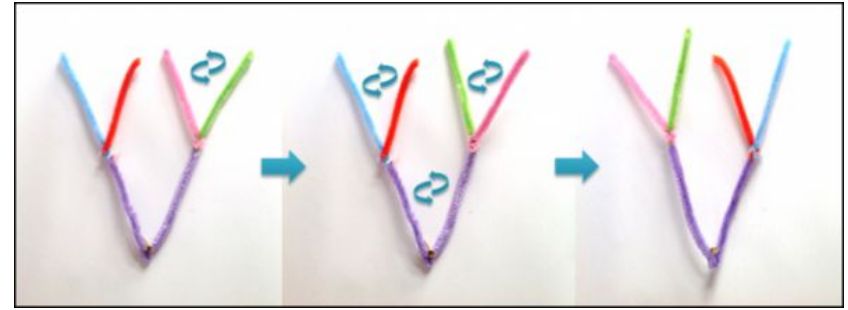


Aspect of phylogeny

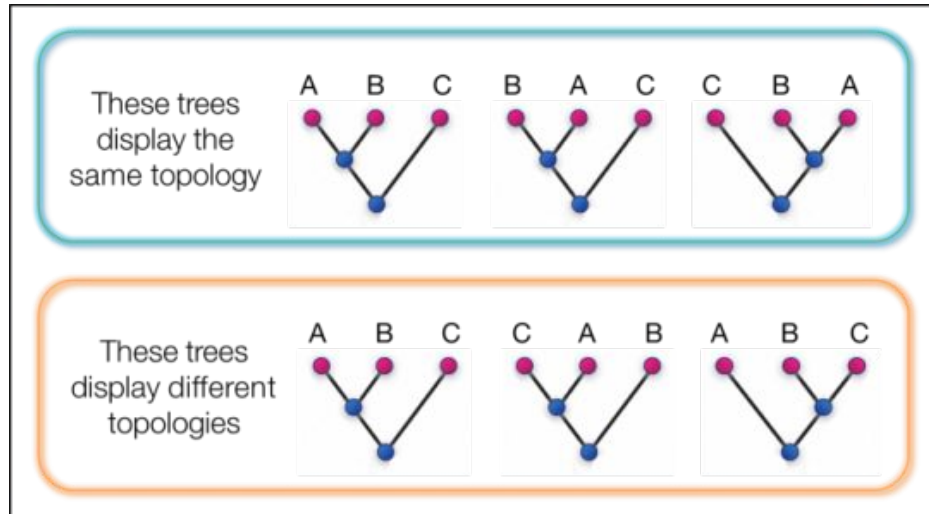
- Topology
- Branches
- Nodes
 - Tips
 - Internal nodes
 - Root



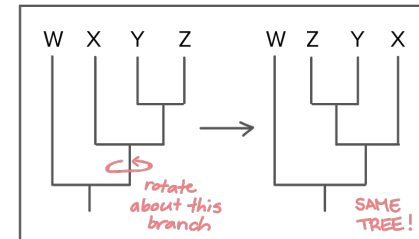
Topology



= branching structure of the tree = indicates patterns of relatedness among taxa



A and B are more closely related to one another than they are to C

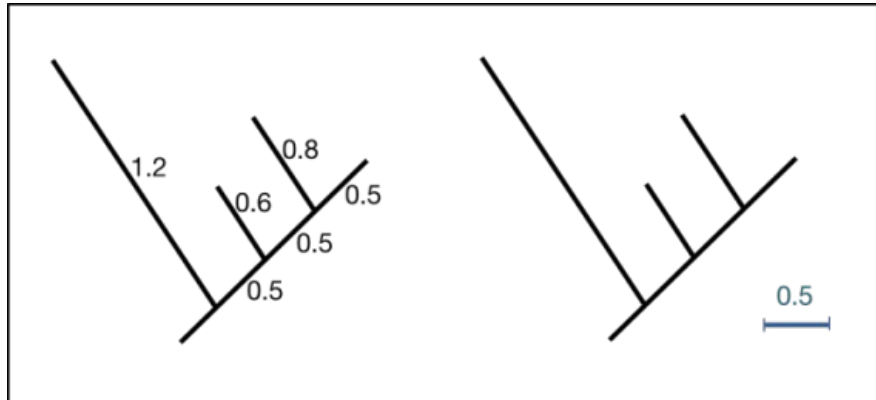




Branches

= path of transmission of genetic information from one generation to the next

-> the longer the branch, the more genetic change (or divergence) has occurred





How to estimate genetic change ?

Human	ATG T TGACTC
Mouse	ATG C TGACTC

$1/10 = 0.1$ substitutions per site



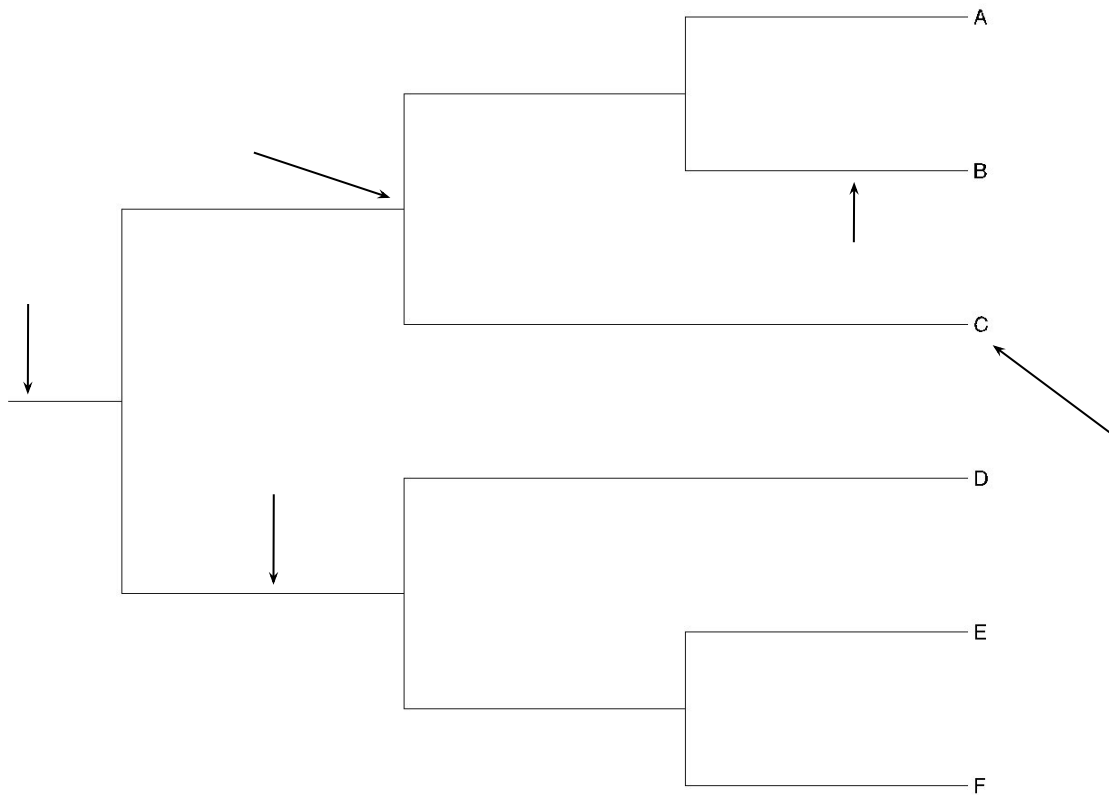
What do we assume here :

- we have observed every substitution that has happened
- every substitution (e.g. from T>C, or A>G) is equally likely to have occurred

-> **evolutionary model**

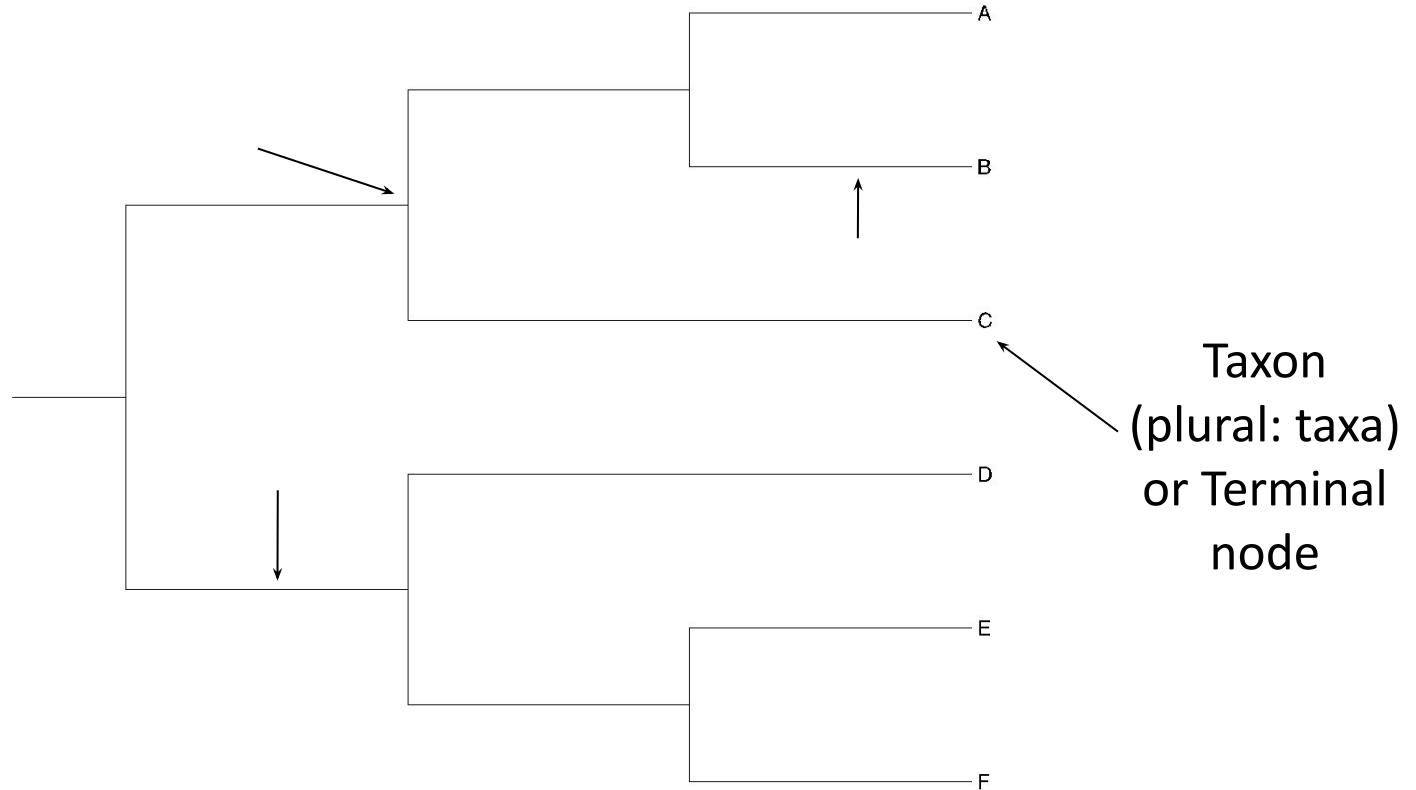


Nodes



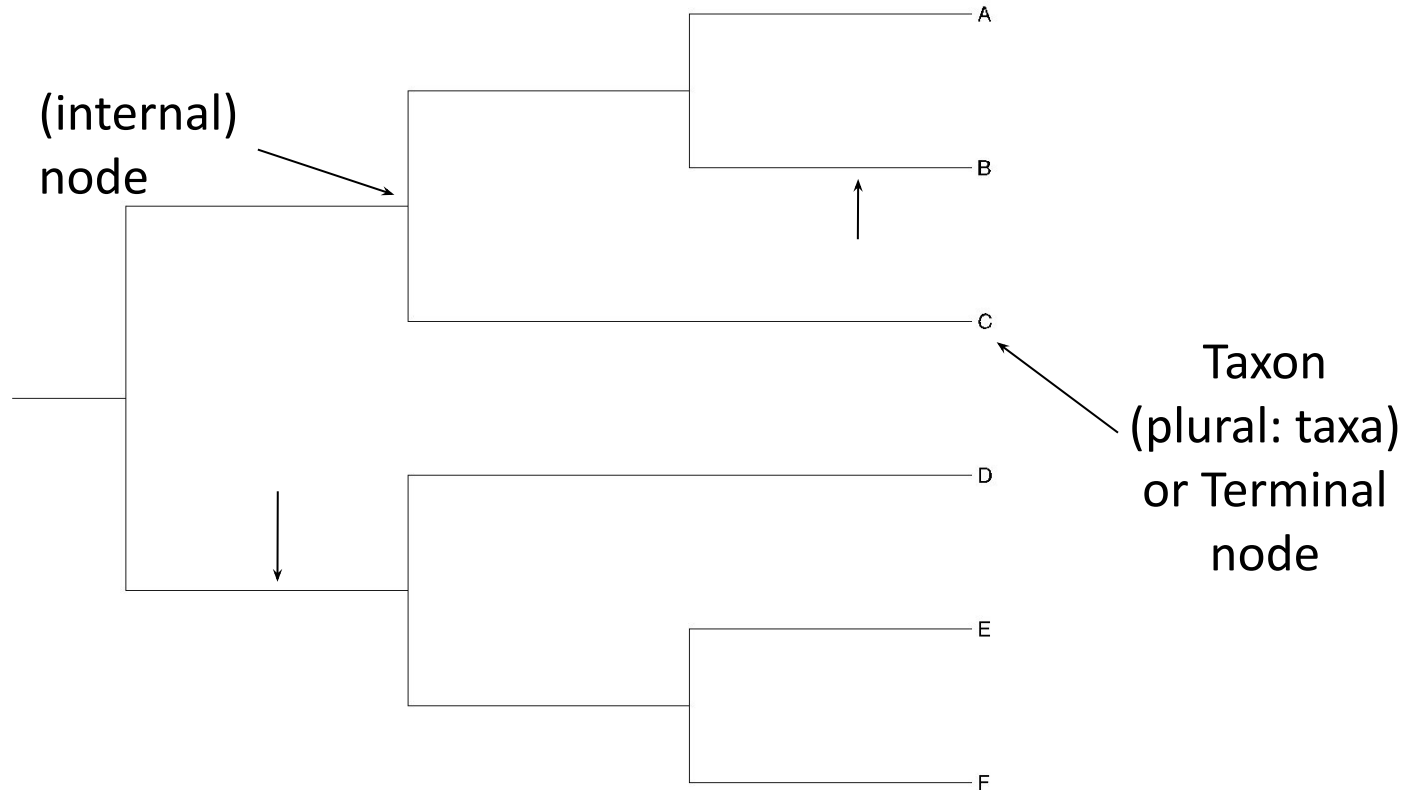


Nodes



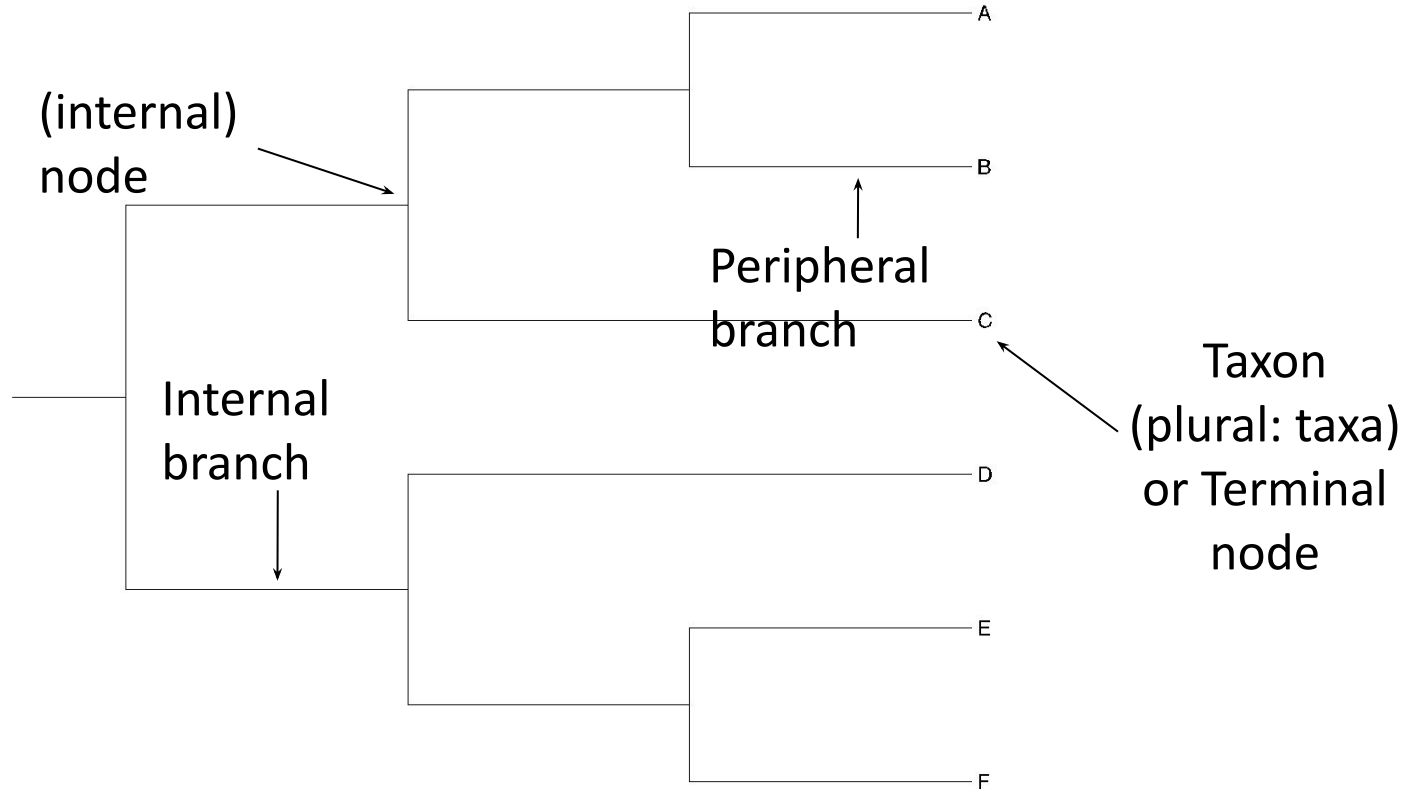


Nodes



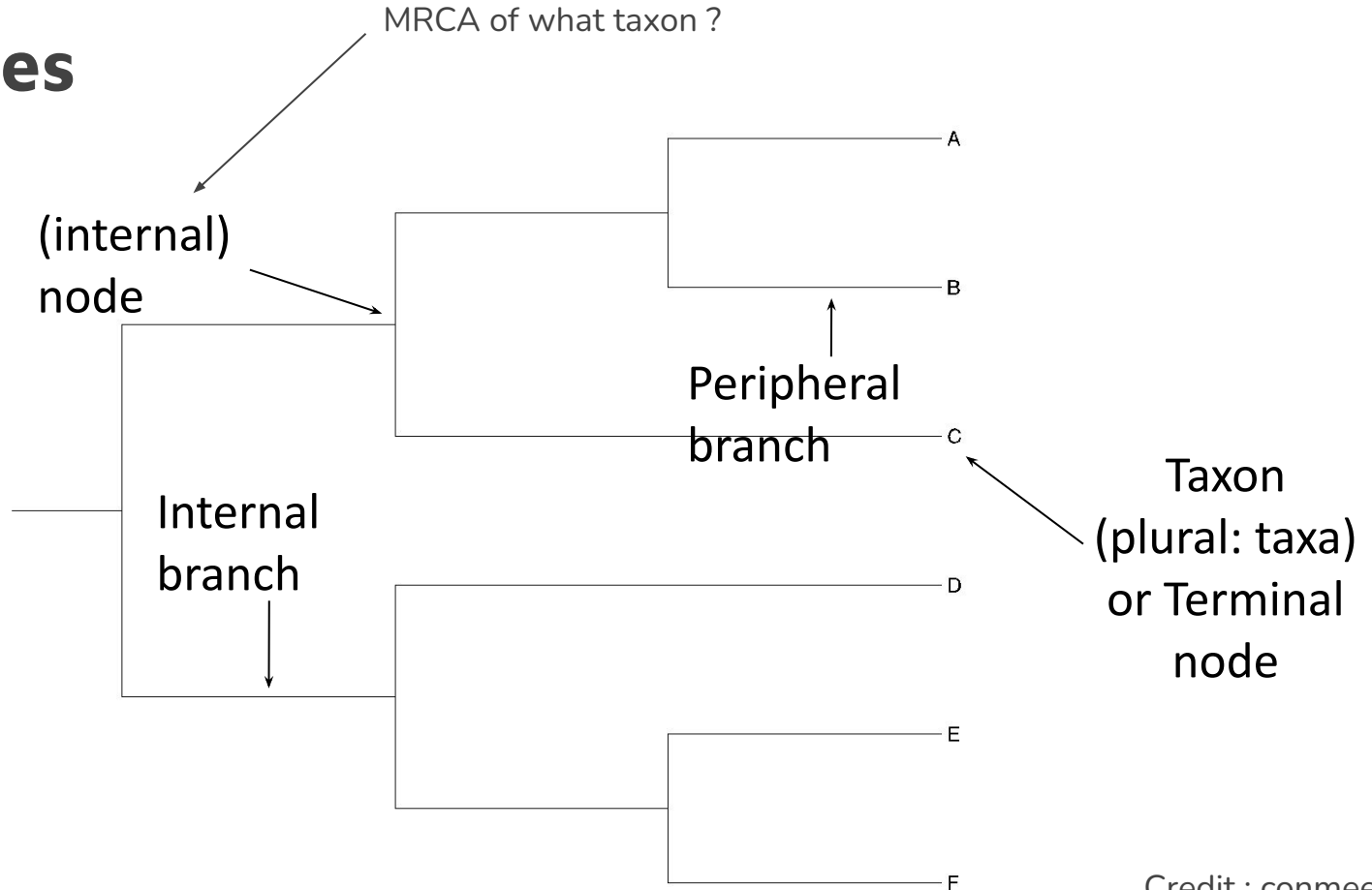


Nodes





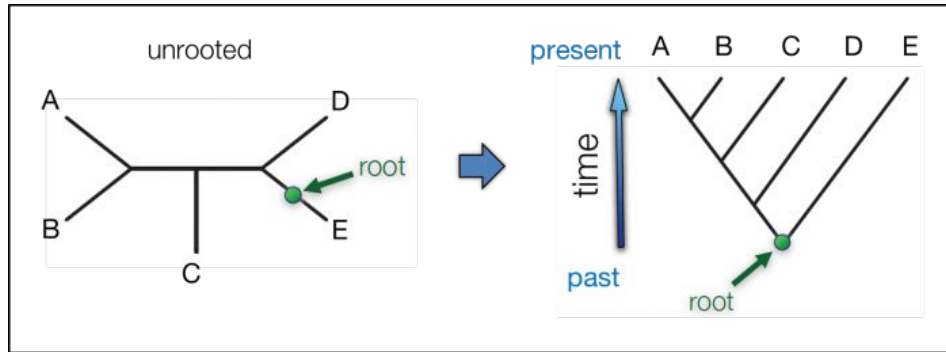
Nodes





Root

= the most recent common ancestor of all of the taxa in the tree

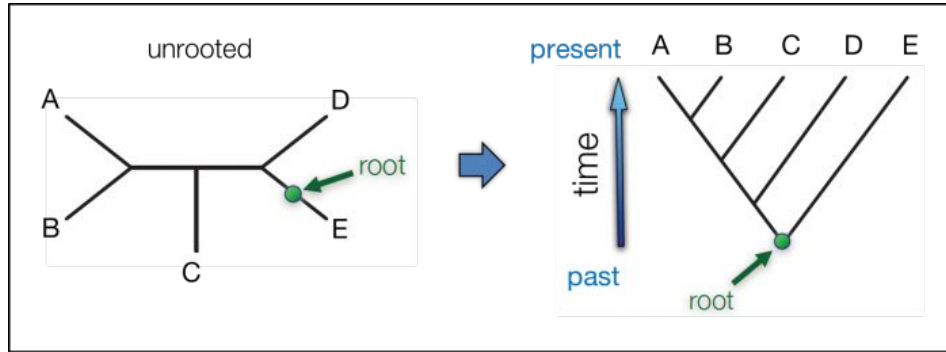


Most phylogenetic reconstruction do not estimate the position of the root



Root

= the most recent common ancestor of all of the taxa in the tree



Most phylogenetic reconstruction do not estimate the position of the root



Rooting a tree affects its meaning

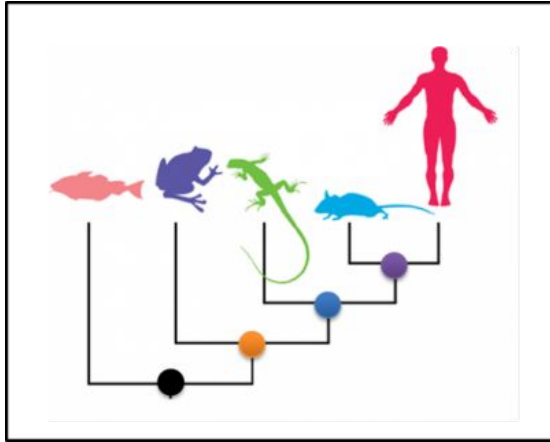
the root tells us the direction of evolution and so affects statements that we make about patterns of relatedness

How to root :

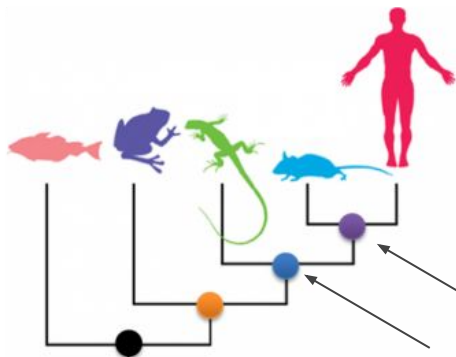
- Outgroups
- Midpoint rooting



Interpreting patterns of relatedness



Are humans (red) more closely related to mice (bright blue) than they are to lizards (green) ?

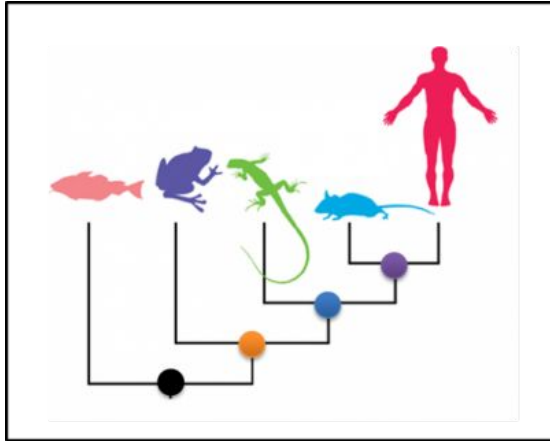


YES

YES



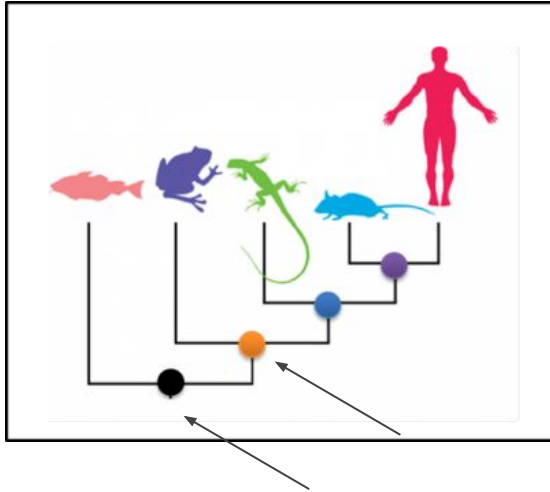
Interpreting patterns of relatedness



Are Frogs (purple) more closely related to lizards (green) than they are to fish (pink) ?



Interpreting patterns of relatedness

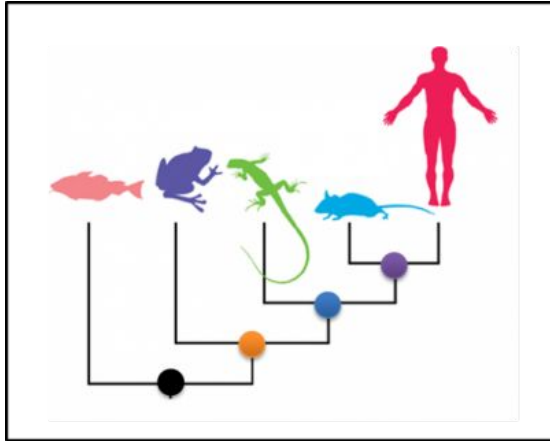


Are Frogs (purple) more closely related to lizards (green) than they are to fish (pink) ?

YES



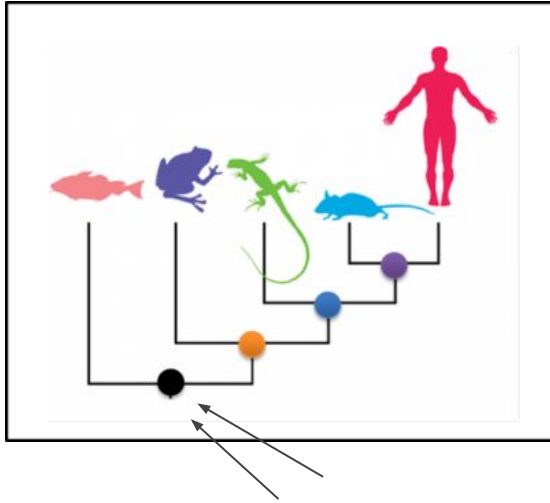
Interpreting patterns of relatedness



Are fish (pink) more related to mice (bright blue) as they are to frogs (purple) ?



Interpreting patterns of relatedness



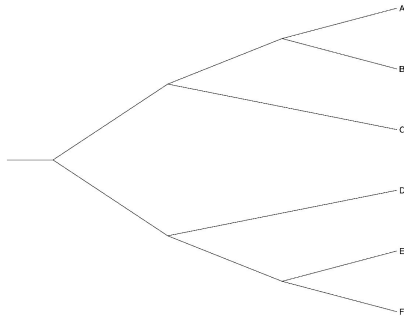
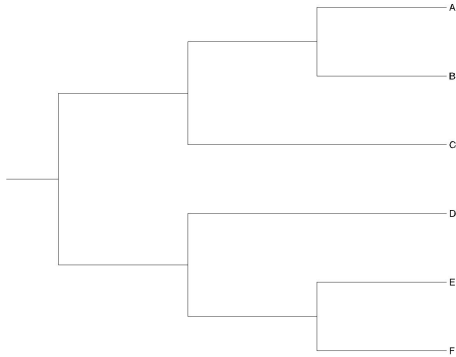
Are fish (pink) more related to mice (bright blue) as they are to frogs (purple) ?

NO, they are equally related

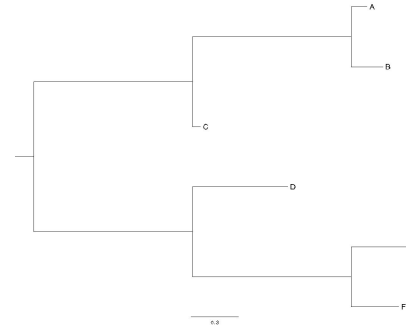


Tree representation

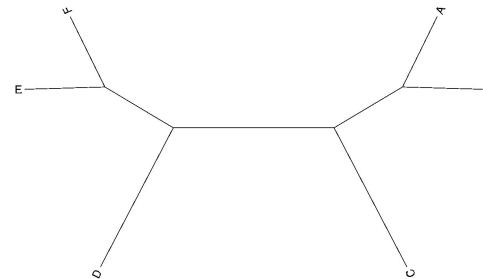
Cladogram: Branch lengths have no meaning
(Sometimes have straight or curved lines)



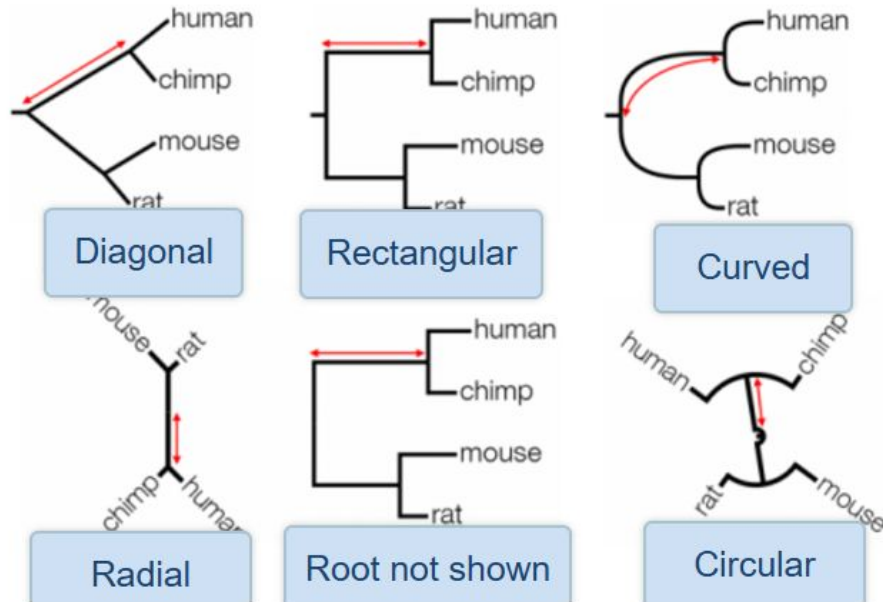
Phylogram: Branch lengths have meaning
(e.g. estimated/expected/observed amount of change or time)



Unrooted: No common ancestor is known or time is not known



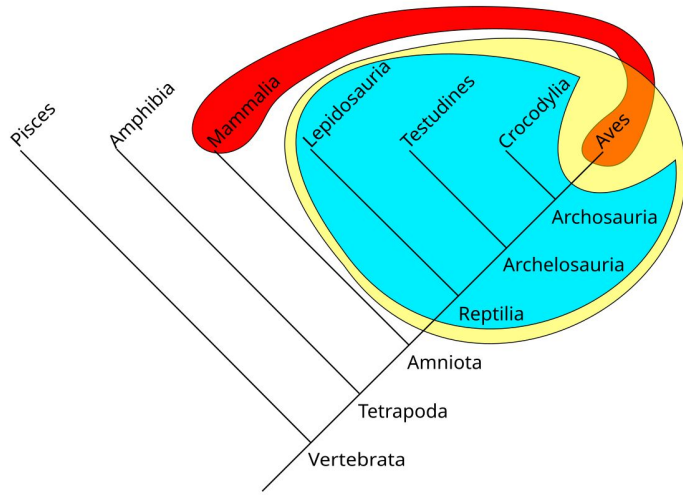
Tree representation





Terminology

Monophyly
Paraphyly
Polyphyly



A monophyletic taxon (in yellow): the group of "reptiles and birds", contains its most recent common ancestor and all descendants of that ancestor.

A paraphyletic taxon (in cyan): the group of reptiles, contains its most recent common ancestor, but does not contain all the descendants (namely Aves) of that ancestor

A polyphyletic "group" (in red): the group of all warm-blooded amniotes (Aves and Mammalia), does not contain the most recent common ancestor of all its members



Terminology

Bifurcating

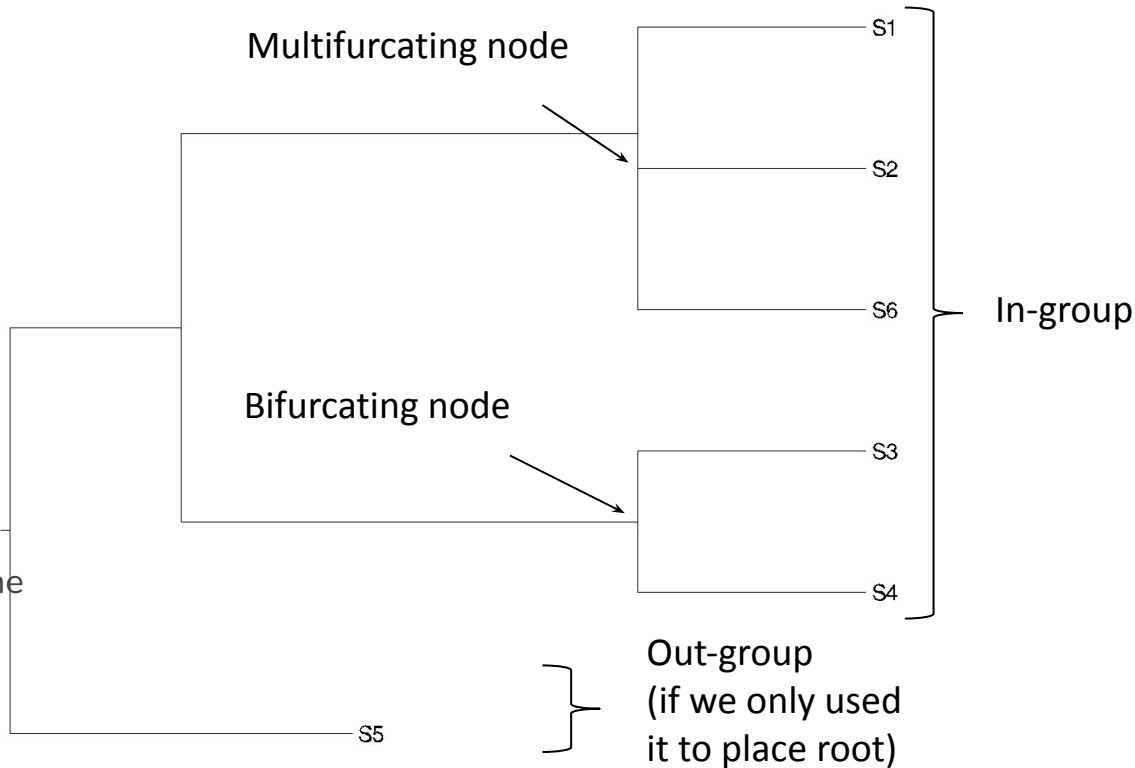
- A node that connects only 2 branches

Multifurcating (polytomy)

- A node in a tree which connects more than two branches

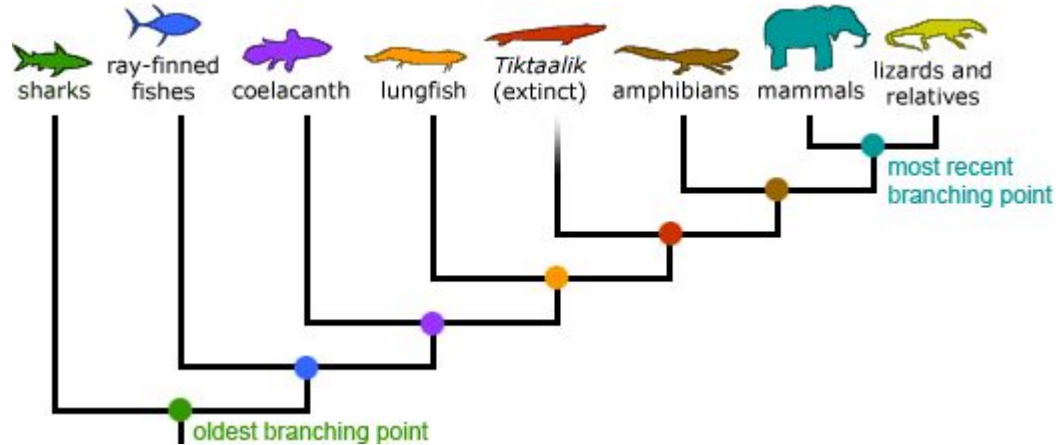
In-group/out-group

- In-group is the set of taxa of interest. Assumed to be monophyletic
- Out-group is a related set of taxa to the in-group, used for rooting



Misconception

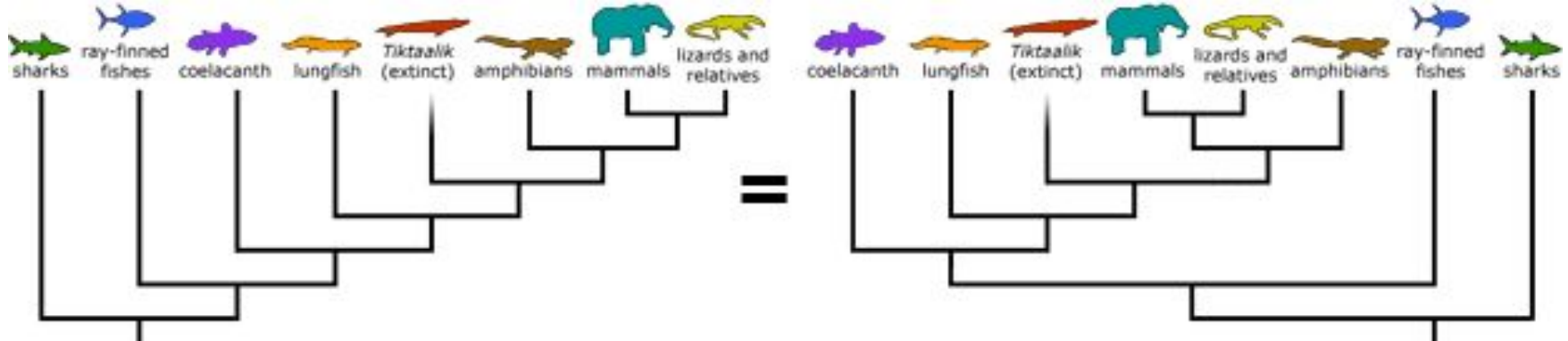
Interpretations rely on the tree being rooted because the root is needed to define the direction of evolution and therefore what is 'more recent' in evolutionary time.





Misconception

If we rotate the branches to change the topology of the tree then the tree still has the same biological meaning and evolutionary relationships





Writing trees on a computer

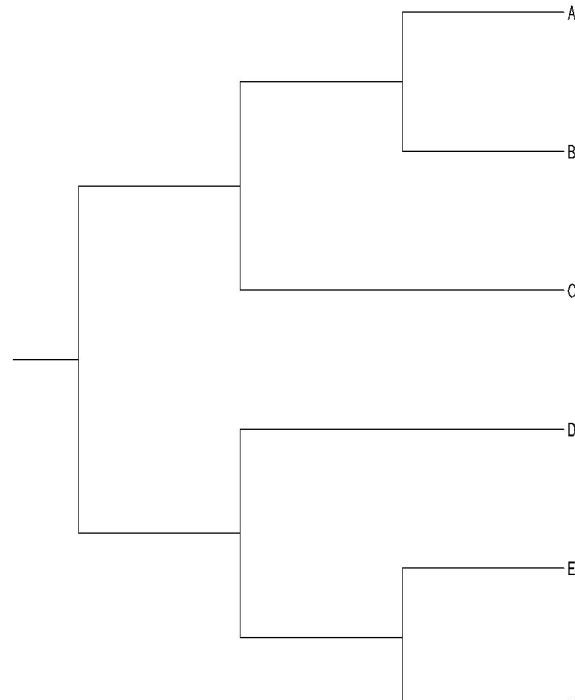
Trees are represented in a text file using parentheses and commas
(newick format)

Every internal node is a comma and the leaf nodes that connect to that internal node have () around them

- This tree is represented as: (((A,B),C),(D,(E,F)))

If we know the lengths of the branches, they are written after each label
or) after a colon

- (((A:0.5,B:0.1):0.2,C:0.5):0.3,(D:0.1,(E:0.2,F:0.2)))





How do we create a phylogeny

Use an optimality criterion to define how we measure the fit of the data to a solution

Tree-search method: how do we decide between possible solutions?

Simplest is parsimony

- Referred to as non-parametric
- The tree that represents the minimum number of character changes between taxa is the optimal solution

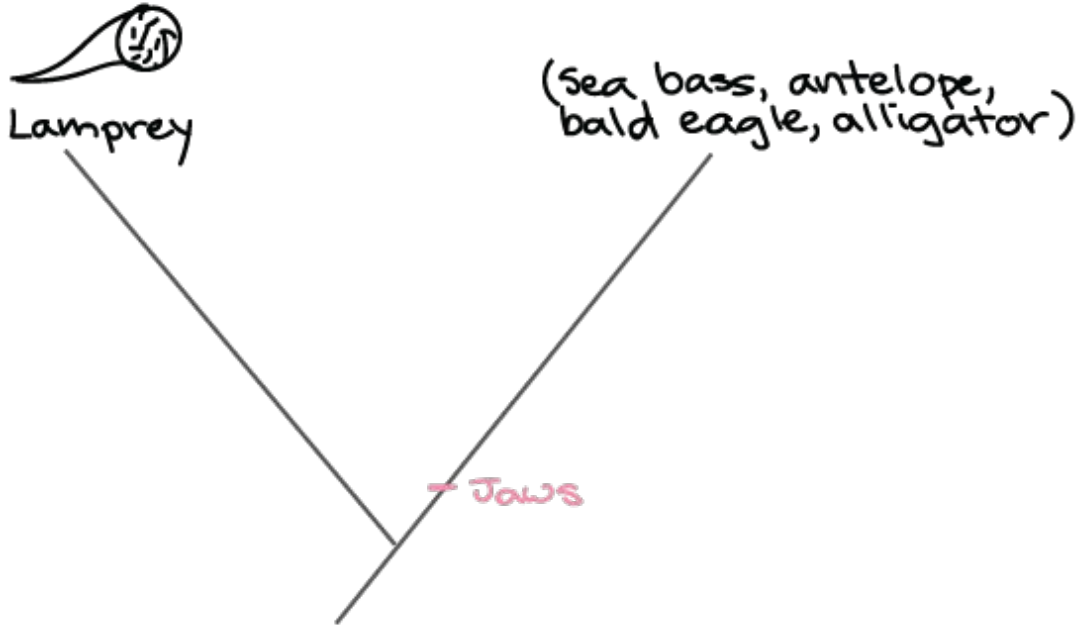


Morphological example

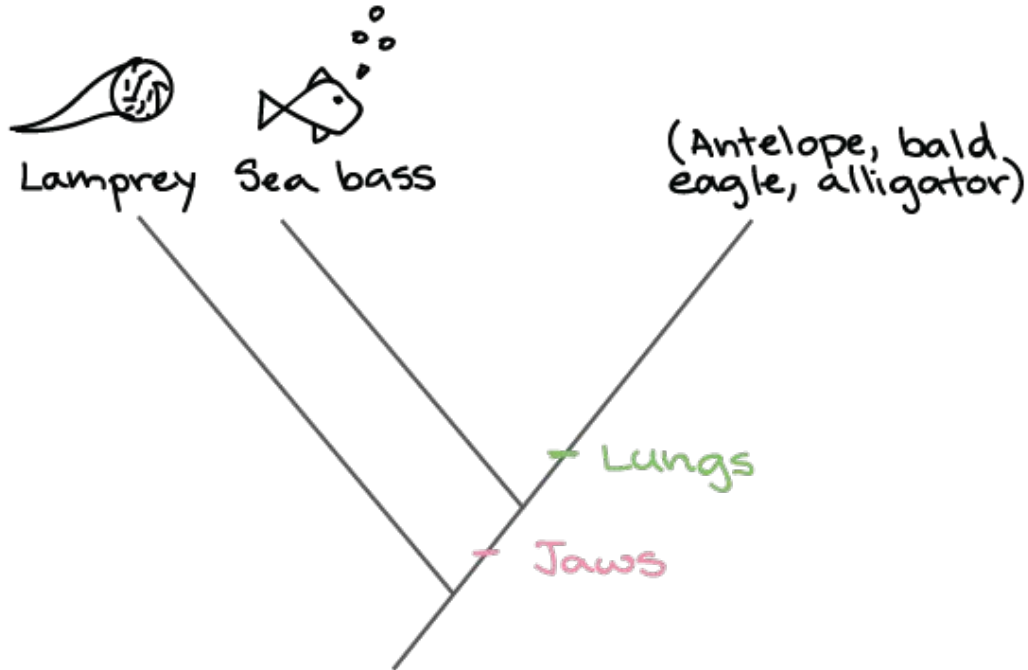


Feature	Lamprey	Antelope	Bald eagle	Alligator	Sea bass
Lungs	0	+	+	+	0
Jaws	0	+	+	+	+
Feathers	0	0	+	0	0
Gizzard	0	0	+	+	0
Fur	0	+	0	0	0

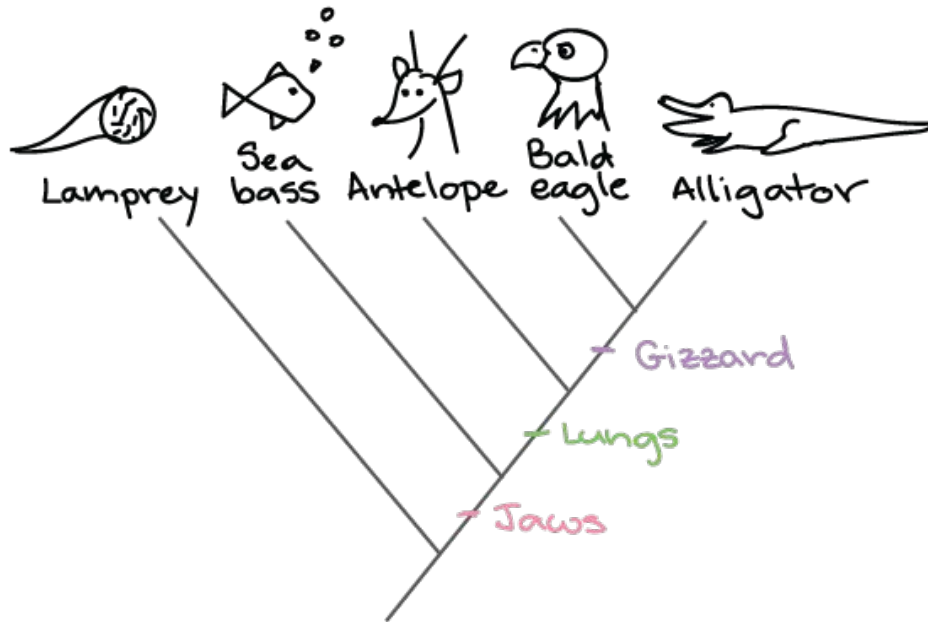
Morphological example



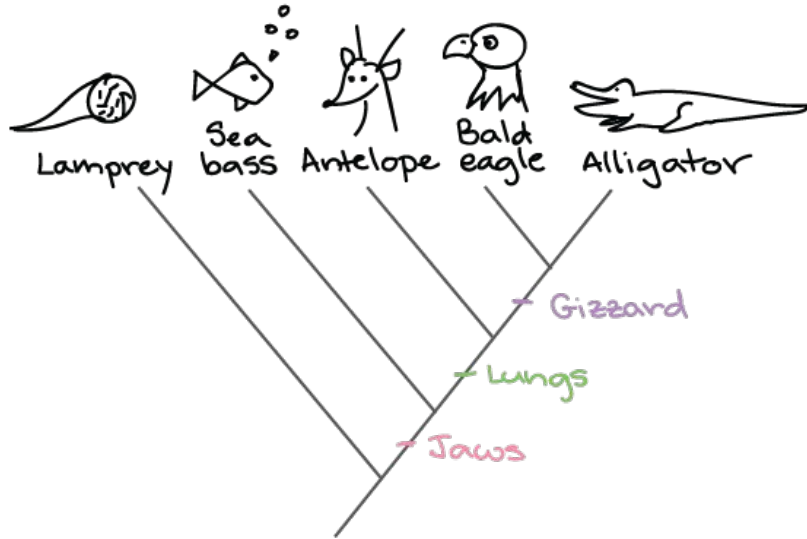
Morphological example



Morphological example

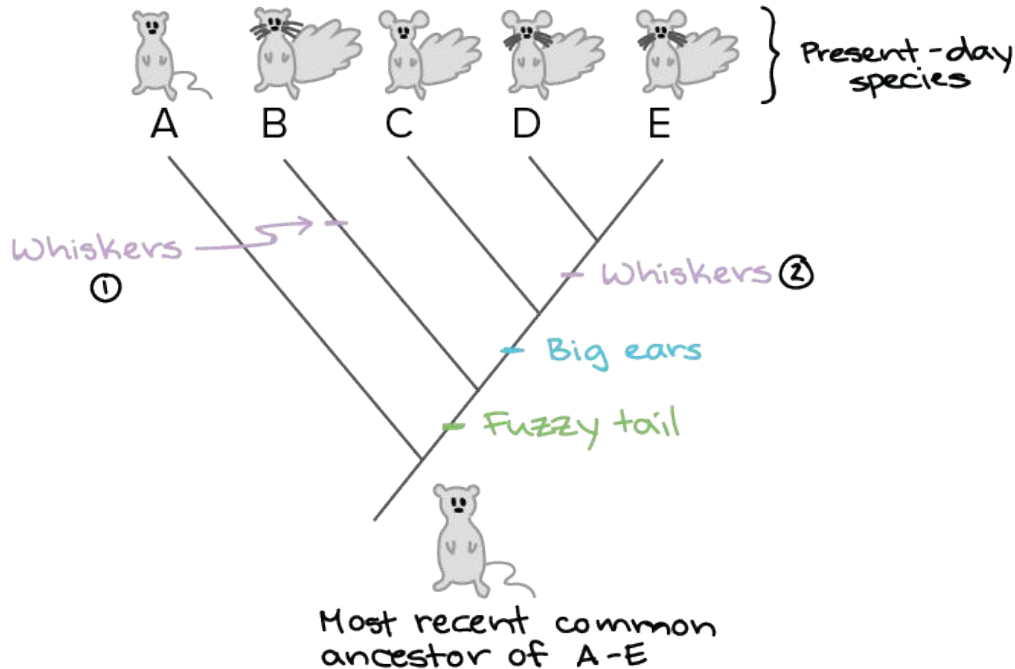


Morphological example



The primary problem with parsimony

CONVERGENT EVOLUTION of WHISKERS



Convergent evolution: the evolution of the same trait multiple times

E.g. wings in bats and birds



Why use molecular data?

Phylogeny is most often undertaken using molecular data

- Nucleotides/DNA sequence (sometimes RNA sequences)
- Amino acids/Protein sequence

Molecular data are :

- Inherited material
- Less sensitive to convergent evolution
- Highly specific

Parsimony also can be used on molecular data

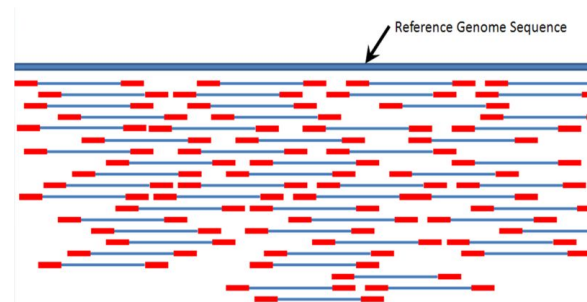
- Use G, T, C, A or amino acids as the characters
- Fine on very closely related isolates (e.g. local transmission cluster separated by a couple of SNPs)

Early tree building used distance based methods

- Semi-parametric
- Get a distance between 2 sequences
- Sequences with shortest distance are most related
- Most often uses UPGMA or neighbour joining method

How to make an alignment for a phylogenetic tree?

- DNA / Amino acids
- MSA (Multiple Sequence Alignment)
 - Clustal, MUSCLE, MAFFT, MAUVE, HMMER...
- Reference Mapping + variant calling
 - BWA, Minimap, Bowtie2
 - Snippy (Freebayes), GATK





UPGMA (sequences)

S1 : TTCAG

S2 : TTCGG

S3 : TTTTG

S4 : TTATG

S5 : AACTG



UPGMA (distance matrix)

		S1	S2	S3	S4	S5
S1 : TTCAG	S1					
S2 : TTCGG	S2					
S3 : TTTTG	S3					
S4 : TTATG	S4					
S5 : AACTG	S5					

Note: count as differences, not as number of characters in common
Dissimilarity is differences/total sites



UPGMA (dendogram)

		S1	S2	S3	S4	S5
S1 : TTCAG	S1	-	0.2	0.4	0.4	0.6
S2 : TTCTG	S2	0.2	-	0.4	0.4	0.6
S3 : TTTCTG	S3	0.4	0.4	-	0.2	0.6
S4 : TTATG	S4	0.4	0.4	0.2	-	0.6
S5 : AACTG	S5	0.6	0.6	0.6	0.6	-

Note: count as differences, not as number of characters in common
Dissimilarity is differences/total sites



UPGMA tree building steps

1. Start with every sequence in its own cluster
2. Select the smallest distance between clusters
 - a. Create a new cluster by joining these two
 - b. Branch lengths are $\text{distance}/2$
3. Get the distance from that cluster to all others
 - a. Use a proportional average distance (i.e. using $\text{size of cluster} \times \text{distance}$)
4. Repeat steps 2 and 3 until complete



UPGMA tree building steps

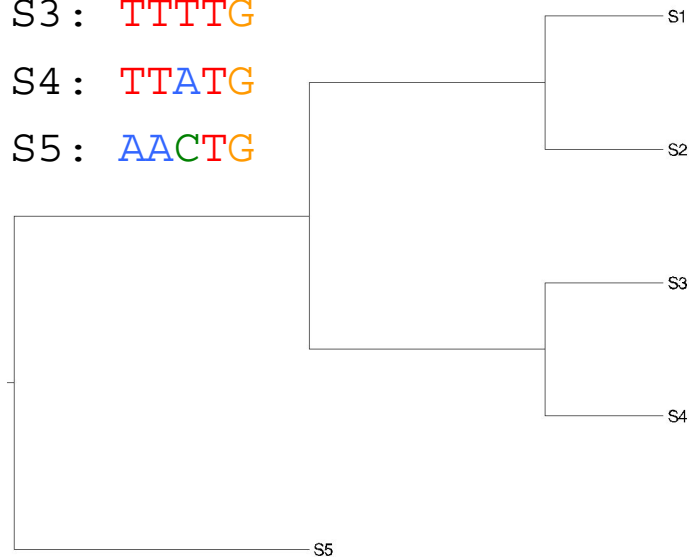
S1: TTCAG

S2: TTCGG

S3: TTTTG

S4: TTATG

S5: AACTG



1. S1-S2 and S3-S4 are both smallest so randomly choose S1-S2
2. New distances to S3 is $(0.4+0.4)/2 = 0.4$
 - a. Repeat for all sequences
3. Get smallest distance and repeat 1 and 2 to build entire tree

	S1	S2	S3	S4	S5
S1	-	0.2	0.4	0.4	0.6
S2	0.2	-	0.4	0.4	0.6
S3	0.4	0.4	-	0.2	0.6
S4	0.4	0.4	0.2	-	0.6
S5	0.6	0.6	0.6	0.6	-



UPGMA tree building steps

Limits

- What about unobserved back mutations over long periods of time?
 - A→G→A in a sequence
 - Underestimated distances
- Are all substitutions equal?
 - Some more likely than others (Transitions vs transversions)
- Need to correct distances
 - Use a model of evolution



BREAK

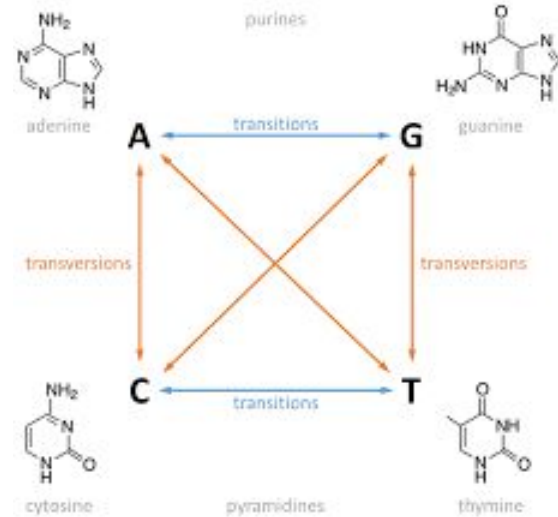


Models of evolution

To account for multiple mutations and weight substitutions based on different factors we need a model of evolution.

These factors include

- Chemical properties
 - E.g. Transition vs transversion
- The relative frequencies of each base over time
 - E.g. GC content
 - Equilibrium frequencies (π)
- Position in the sequence
 - E.g. Hypervariable vs core
 - Rate heterogeneity/site-specific variation (γ)



For distance approaches (e.g. UPGMA), we need to correct the distance based on these factors



Simplest model

The Jukes Cantor model states:

- all nucleotide changes have equal probabilities
 - (A→C/G/T = 0.25)
- All base frequencies are the same
 - $\pi_a = \pi_c = \pi_g = \pi_t$

$$d_{jc} = -3/4 \ln(1 - 4/3 D)$$

- d_{jc} is the corrected distance
- D is the original (uncorrected) distance
- Works when distances are < 0.75

Distance between S2 & S3

S1: TTCAG

S2: TTCGG

S3: TTTTG

S4: TTATG

S5: AACTG

$$D = 2/5 = 0.4$$

$$d_{jc} = -3/4 \ln(1 - 4/3 * 0.4)$$

$$d_{jc} = 0.4714564946$$



Different models of evolution

Jukes-Cantor (often called JC69)

- Rates are the same
- Frequencies are the same

K80 (also called Kimura 2 parameter (K2P))

- Transitions (A-G or C-T) different rate to transversions
- Frequencies are the same

F81

- Rates are the same
- Frequencies can vary from 0.25 (as they are in JC69)

HKY85

- Combination of the above



Rate heterogeneity

```
RLA0_METVA  --MIDAKSEHKIAPWKIEEVNALKELIKSANVIALIDMMEVPAVQLQEIRDK
RLA0_METJA   ---METKVKAHVAPWKIEEVKTLKGLIKSKPVVAIVDMMDVPAPQLQEIRDK
RLA0_PYRAB   -----MAHVAEWKKKEVEELANLIKSYPVIALVDVSSMPAYPLSQMRRL
RLA0_PYRHO   -----MAHVAEWKKKEVEELAKLIKSYPVIALVDVSSMPAYPLSQMRRL
RLA0_PYRFU   -----MAHVAEWKKKEVEELANLIKSYPVVALVDVSSMPAYPLSQMRRL
RLA0_PYRKO   -----MAHVAEWKKKEVEELANIIKSYPVIALVDVAGVPAYPLSKMRDK
RLA0_HALMA   MSAESERKTETIPEWKQEEVDIVEMIESYESVGVVNIAGIPSRQLQDMRRD
RLA0_HALVO   MSESEVRQTEVIPQWKREEVDLVDLIESYESVGVVGVAGIPSRQLQSMRRE
RLA0_HALSA   MSAAEQRTTEEVPEWKRQEVAVELVDLLETYDSVGVVNVGTGIPSKQLQDMRRG
RLA0_THEAC   -----MKEVSQQKKELVNEITQRIKASRSVAIVDTAGIRTRQIQDIRGK
RLA0_THEVO   -----MRKINPKKKEIVSELAQDITKSKAVAIVDIKGVTRQMODIRAK
RLA0_PICTO   -----MTEPAQWKIDFVKNLENEINSRKVAAIIVSIKGLRNNEFQKIRNS
```



Distance matrices and models of evolution

For distance methods we can directly modify the distance between two sequences

- Transitions/transversions
- Base frequency

But we can't incorporate position heterogeneity in the sequence

Can only handle simple models



Parametric methods of phylogenetic inference

- Parametric methods take a column in an alignment and calculate the optimality criterion per position in alignment
 - Positions are independent
- There are two main parametric methods of phylogenetic inference:
 - Maximum likelihood
 - Bayesian analysis



Parametric methods of phylogenetic inference

- Both methods:
 - Search tree space to find the best tree
 - Require an explicit model of evolution
 - Usually GTR
 - Can incorporate rate heterogeneity

GTR (generalised time-reversible) model :

- Reversible : model takes into account substitutions that can occur in both directions
 - $A \leftrightarrow T$
- Generalized : allows for different substitution rates between each pair of nucleotides
 - $A \rightarrow T$ have rates different $A \rightarrow G$
- All rates and frequencies free to vary
- All previous models are submodels of the GTR



Tree space searching

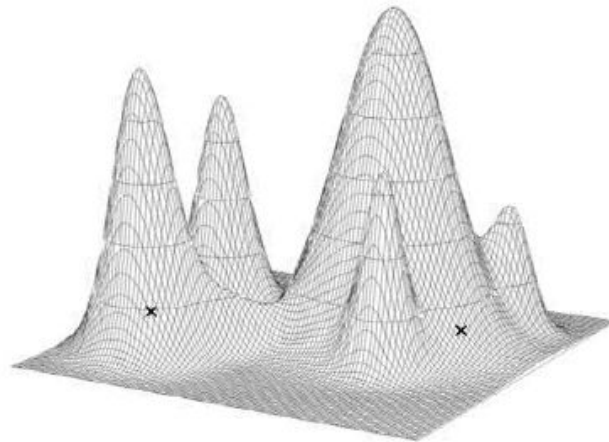
You are randomly in the world and told to find the highest mountain (the global maximum)

You walk in a random direction until you find a section that is sloped upwards

You continue to walk upwards until every direction around them is a downwards slope

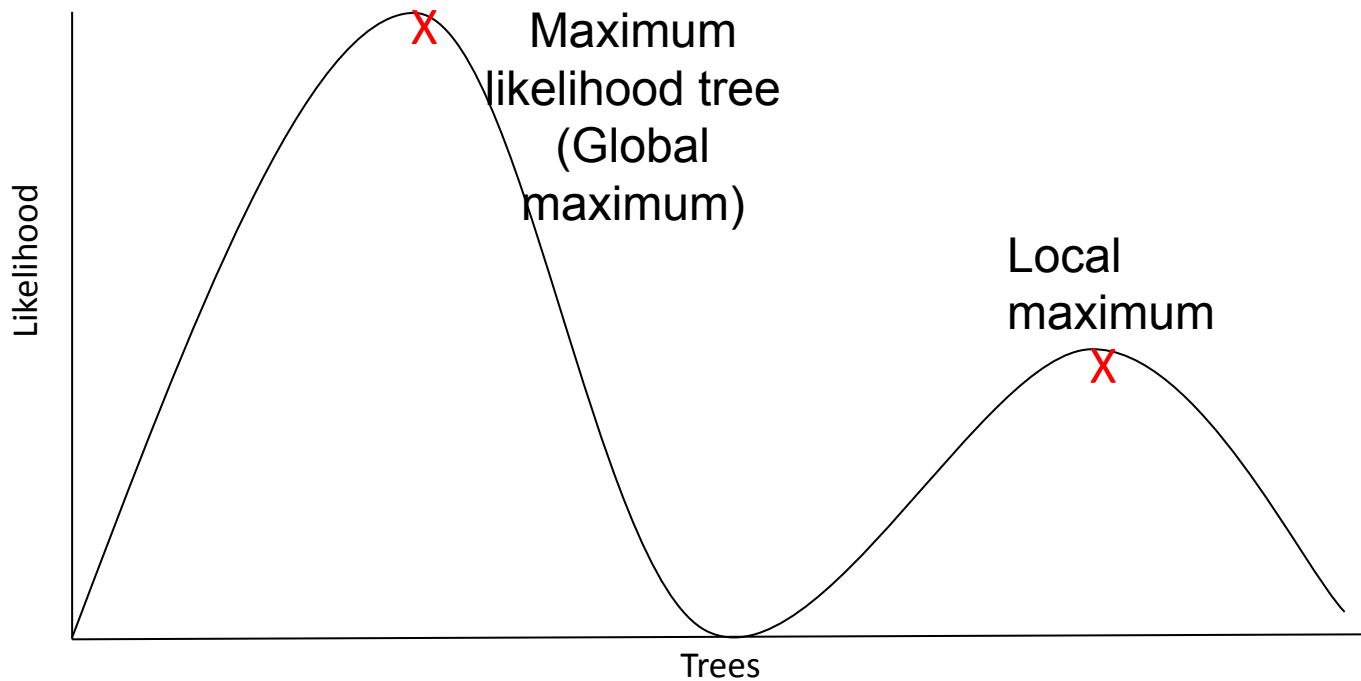
You conclude that since you are at the highest point, you must be on Everest

-> the highest position you stand at has the maximum likelihood of being Everest, given the data and starting point





Tree space searching





Tree space searching

Computer programs will implement different strategies to try and get around this

- Multiple starting points
- Multiple searches at once; can switch between searching chains
- Allow large and small rearrangements
- Allow some steps backwards to try improve score



Maximum Likelihood (ML) trees

- A tree topology is proposed
- A likelihood score (= how well it predicts the observed data) is calculated for each position and added up to get an overall likelihood score for the data for a given tree topology
- Parameters to estimate:
 - Shape of the tree
 - Branch length
 - Model parameters
 -
- You calculate : “how likely is it that the sequences I observe evolved this way?”

$$L(\theta \mid D) = P(D \mid \theta)$$

The likelihood $L(\theta \mid D)$ is the probability of observing data under a model with a specific parameter.



Maximum Likelihood (ML) trees

- Searches tree space and tries to find the tree that has the maximum likelihood of generating the given data
 - Compare topologies through optimising variables for each to fit data
 - = the tree (and model settings) that gives the highest probability of producing the observed sequences.
- Example programs: RAxML, IQ-TREE, PAUP*, PhyML



Bayesian trees

Use Bayes' Theorem, which combines:

- Your observed data (DNA or protein sequences),
- A model of evolution
- Your prior beliefs (called "priors") — like what trees or model settings you thought were more likely before seeing the data.

You calculate the **posterior probability of the phylogenetic tree**, i.e. the probability of the tree given the observed data

the posterior probability of θ given D

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

the likelihood, which is the probability of the data given the parameter θ .

the prior probability of θ , i.e. what we know or assume about θ before observing the data.

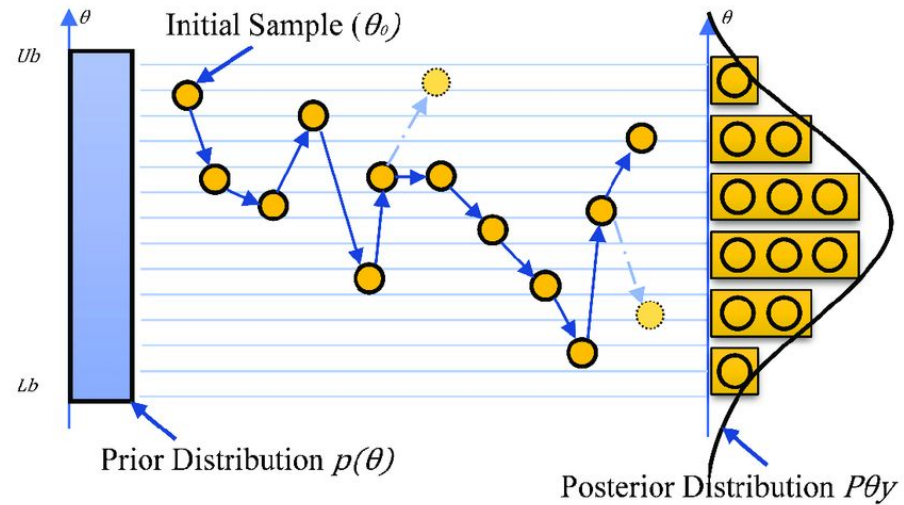
the total probability of the data

Bayesian inference calculates the posterior probability $P(\theta | D)$, which is the probability of a parameter θ after taking priors and knowledge into account.

Bayesian trees

- Start with random tree.
- Propose small changes (e.g. change a branch).
- Use MCMC (Markov Chain Monte Carlo) to:
 - Keep good changes that increase probability.
 - Sometimes accept worse ones to explore other options.
- Repeat many thousands of times.
- After it settles down, you collect the trees, and:
- Count how often each branch or tree shape shows up.
- That becomes your Bayesian posterior tree.

Example programs: MrBayes, BEAST



Bayesian trees

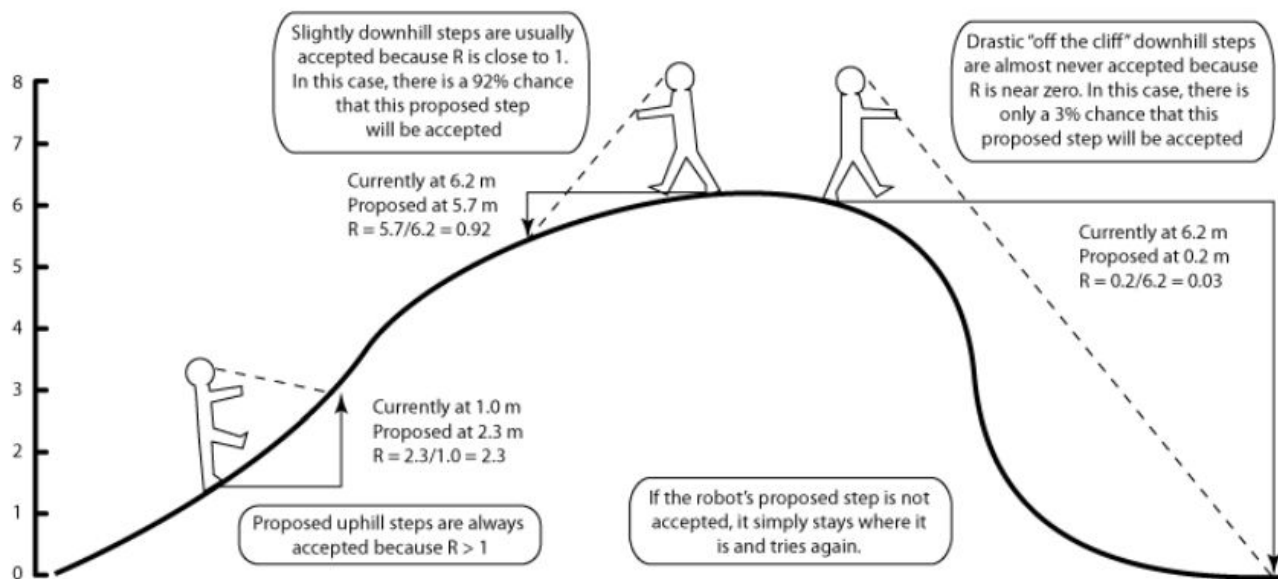


Illustration of MCMC method process (Lewis, 2011)



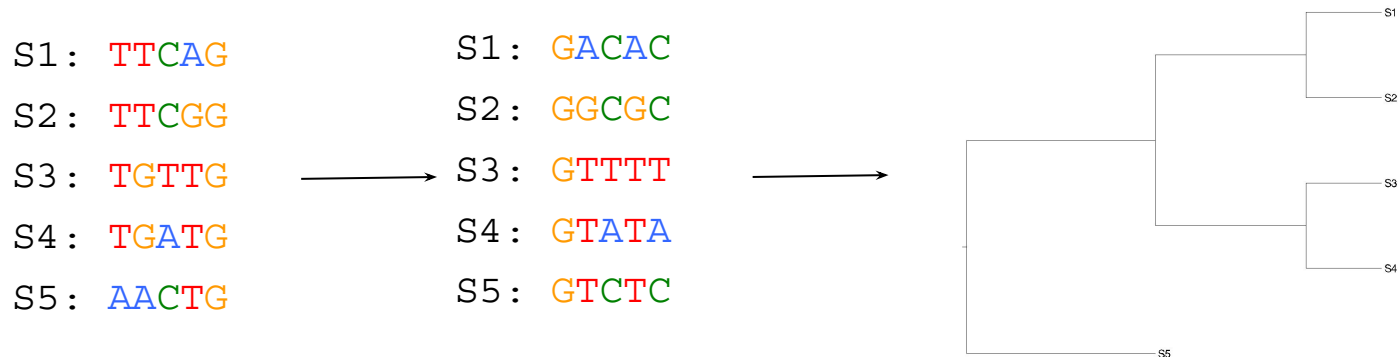
Difference between ML and Bayesian

- ML method: Finds the single best tree that makes your data most likely.
- Bayesian method: Looks at all possible trees and gives probabilities to each one, based on how well they explain the data.



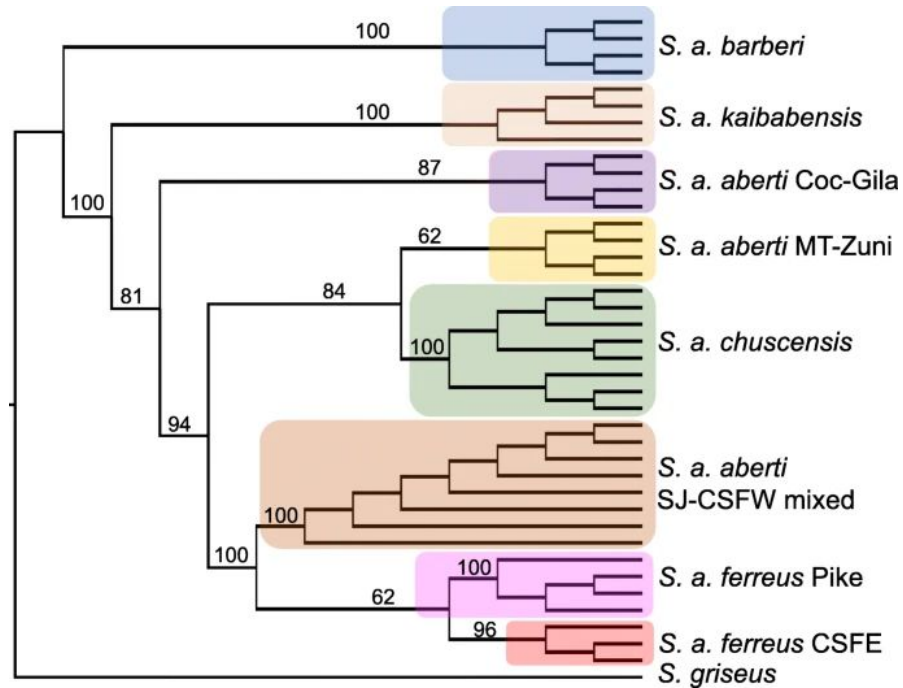
Bootstrapping

- Bootstrapping is a method to test the reliability of an inferred tree
- Algorithm (for an alignment of M taxa and N columns):
 - Create a new alignment of length N from original, sampling columns at random with replacement
 - Create a new topology for this using the same method
 - Compare topology to original
 - Every branch that is the same is given a score of 1, any that are not present are given a score of 0
- Repeat hundred(s) of times and report as a percentage found for each branch



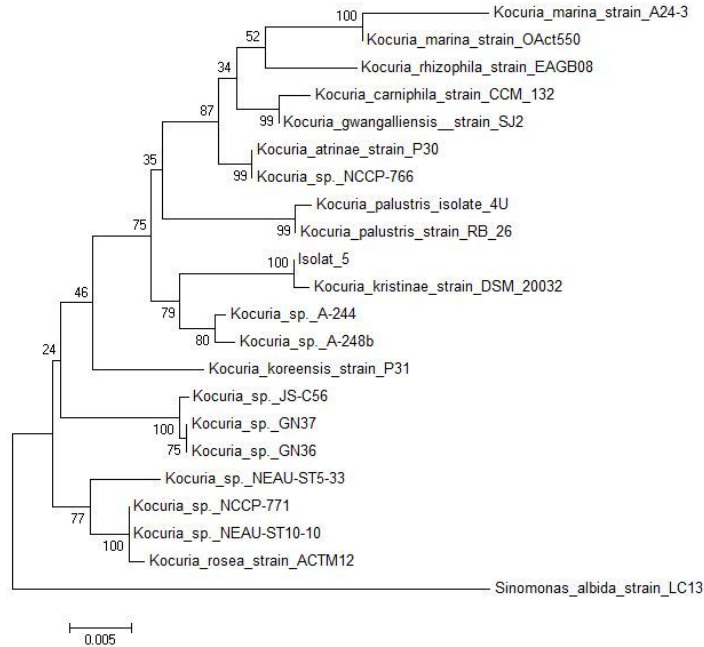


Example



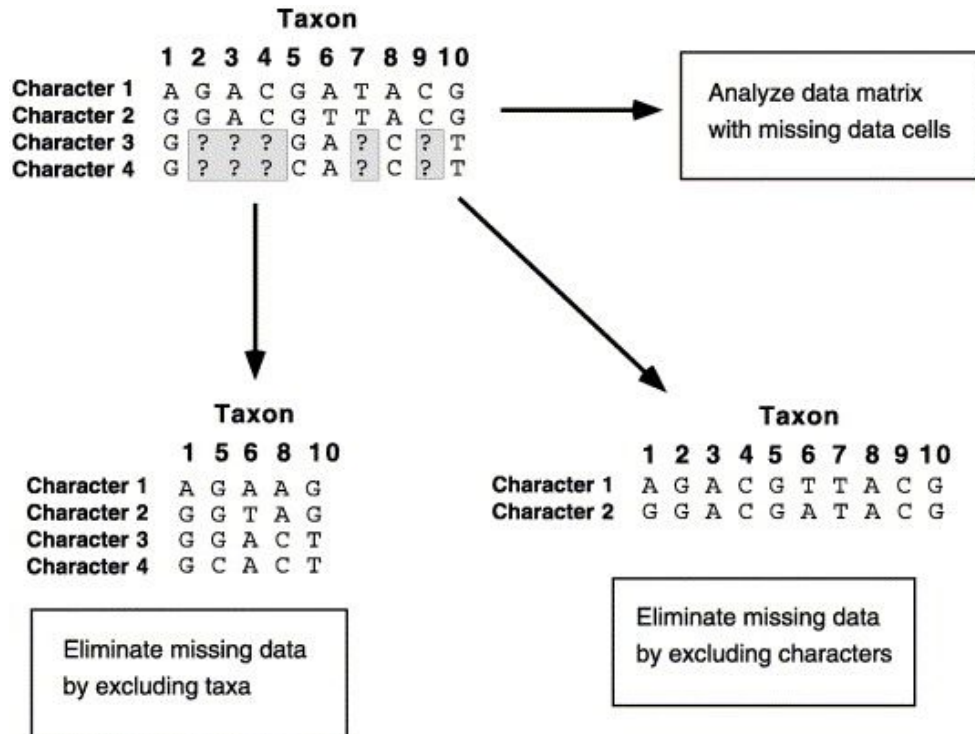


Example

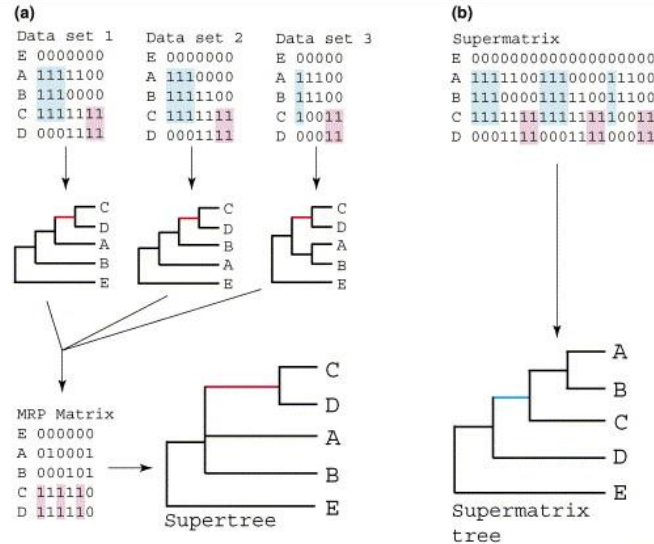


https://www.researchgate.net/post/How_do_I_deal_with_Bad_Bootstrap_Value

Handling missing data

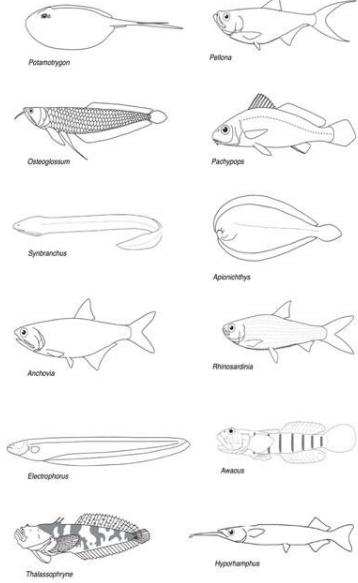


Supertree and supermatrix

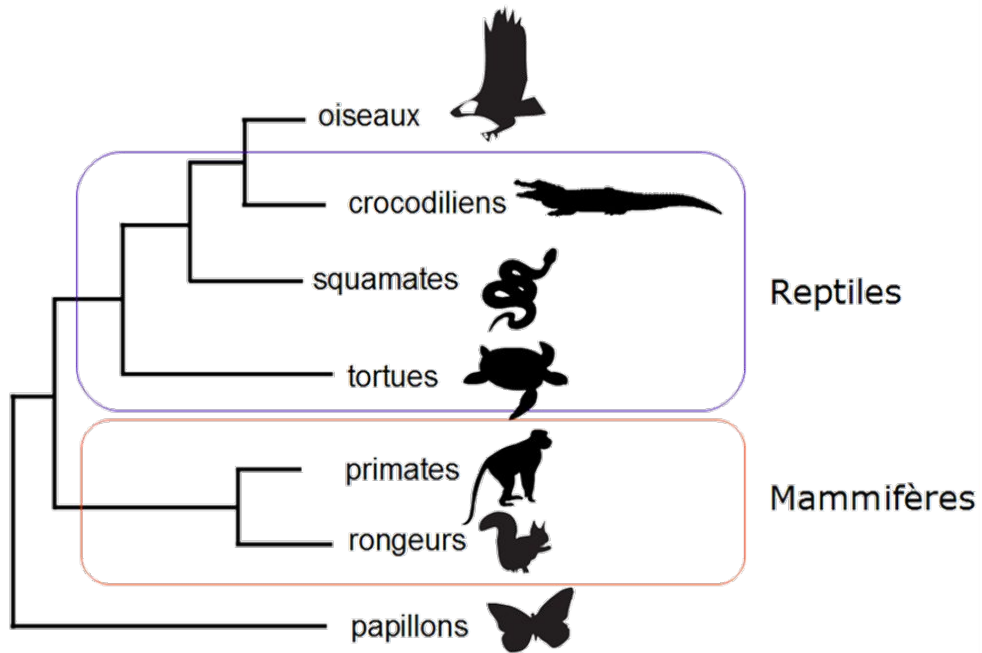


TRENDS in Ecology & Evolution





Reptiles





Mat & Met article

A phylogenetic tree was drawn using multilocus sequence typing (MLST) analysis as described by Duan, et al [5]. The housekeeping genes, *adk* (adenylate kinase), *argA* (amino acid acetyltransferase), *aroA* (3-phosphoshikimate 1-carboxyvinyltransferase), *glnA* (glutamine synthetase), *thrA* (aspartokinase-homoserine dehydrogenase I), *tmk* (thymidilate kinase), and *trpE* (anthranilate synthase component I), were picked from the aforementioned genome sequences. A dendrogram was generated from the concatenated gene sequences via the neighbor joining (NJ) method.

We extracted 16S rDNA sequences from the aforementioned genome sequences and aligned them using MAFFT [21]. These sequences were used to construct neighbor-joining trees using MUSCLE [22].



Mat & Met article

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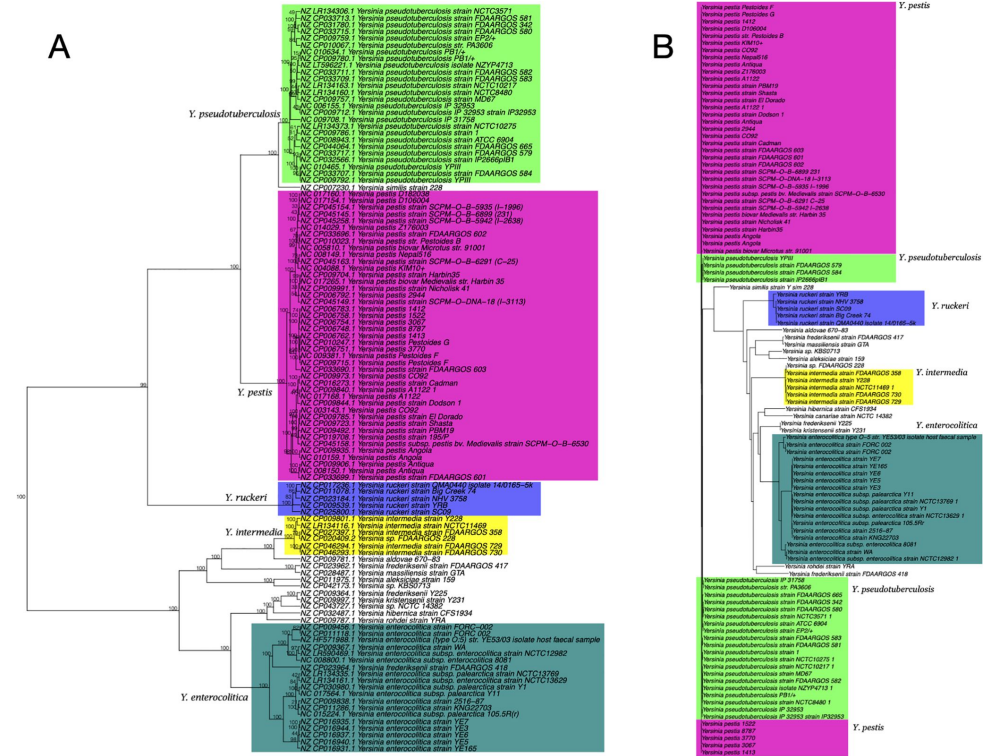


Mat & Met article

Phylogenetic trees of closely related bacterial species and subspecies based on frequencies of short nucleotide sequences

Yoshio Nakano , Yusaku Domon, Kenji Yamagishi

Published: April 20, 2023 • <https://doi.org/10.1371/journal.pone.0268847>





Mat & Met article

For nucleotide as well as amino acid sequences, we performed multiple alignments with clustalw v2.1 (Larkin et al. [2007](#)) followed by maximum likelihood-based phylogenetic inferences with PhyML v3.3.20190909 (Guindon et al. [2010](#)). We assumed a GTR substitution model for nucleotide sequences and an LG substitution model for amino acid sequences, with gamma-distributed substitution rates. The other PhyML parameters were left to their default value.

Source : <https://pmc.ncbi.nlm.nih.gov/articles/PMC7859469/#Sec11>



Mat & Met article

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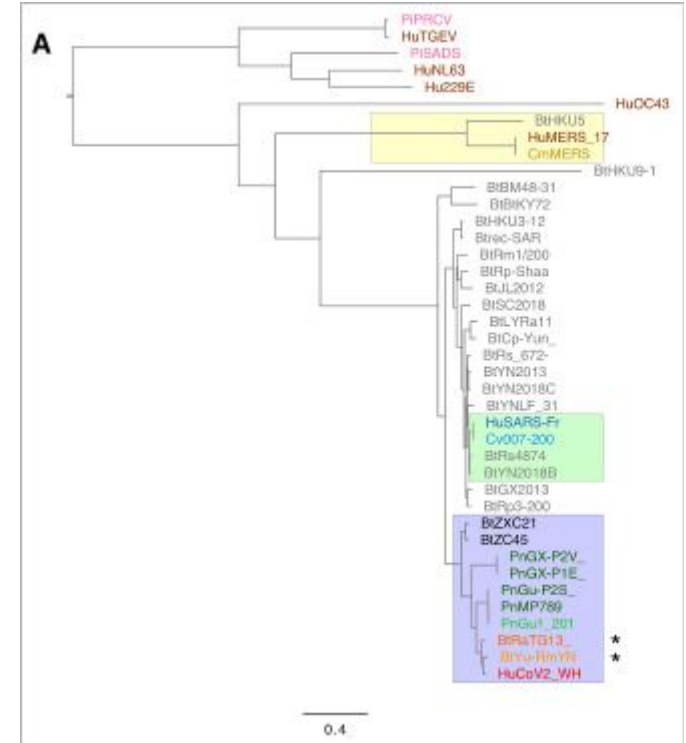
Source : <https://pmc.ncbi.nlm.nih.gov/articles/PMC7859469/#Sec11>



Mat & Met article

Tracing the origins of SARS-COV-2 in coronavirus phylogenies: a review

[Erwan Sallard](#)¹, [José Halloy](#)², [Didier Casane](#)^{3,4}, [Etienne Decroly](#)^{5,#}, [Jacques van Helden](#)^{6,7,✉,#}





Mat & Met article

In BEAST, rates were modelled using the GTR substitution model of evolution and variation among sites was simulated using a discrete gamma distribution with four rate categories. We further assumed a lognormal relaxed clock to allow variation in rates among branches in the tree. The tree was calibrated using tip dates only with sample time span ranging from October 1996 to December 2009. For each analysis we ran four independent chains in which samples were drawn every 5,000 MCMC steps from a total of 50,000,000 steps, after a discarded burn-in of 5,000,000 steps.

Source : <https://pmc.ncbi.nlm.nih.gov/articles/PMC4432642/#sec6>



Mat & Met article

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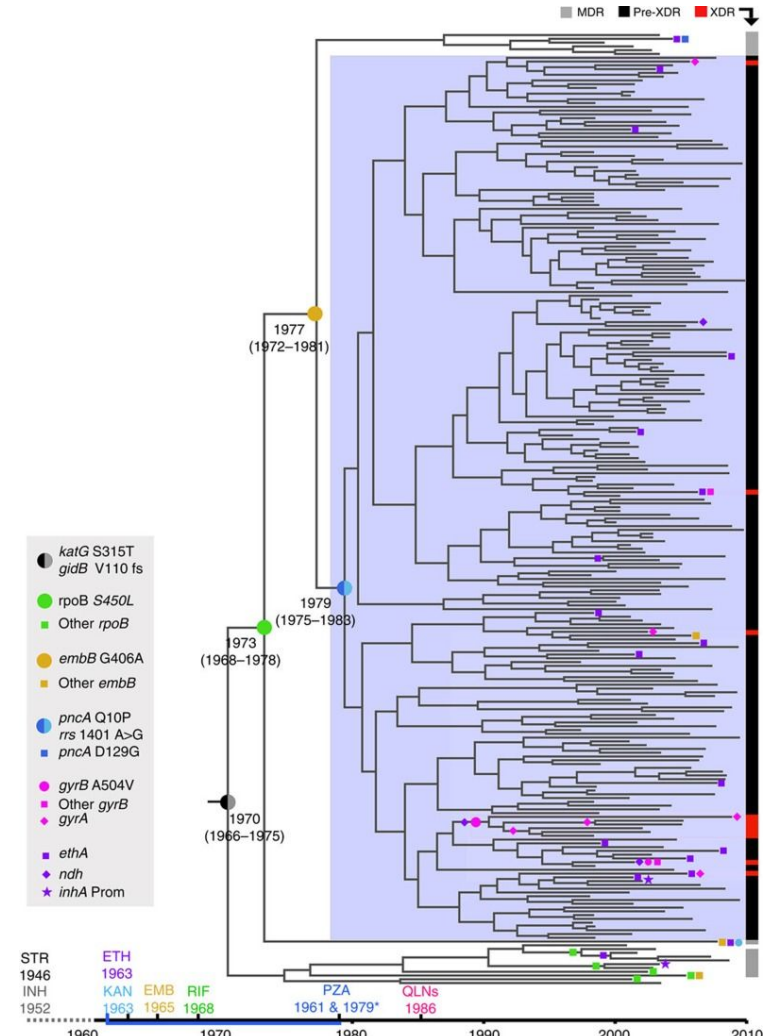
Source : <https://pmc.ncbi.nlm.nih.gov/articles/PMC4432642/#sec6>

Mat & Met article

► Nat Commun. 2015 May 11;6:7119. doi: [10.1038/ncomms8119](https://doi.org/10.1038/ncomms8119)

Four decades of transmission of a multidrug-resistant *Mycobacterium tuberculosis* outbreak strain

[Vegard Eldholm](#)^{1,a}, [Johana Monteserin](#)^{2,*}, [Adrien Rieux](#)^{3,*}, [Beatriz Lopez](#)², [Benjamin Sobkowiak](#)³, [Viviana Ritacco](#)², [Francois Balloux](#)^{3,b}



Source : <https://pmc.ncbi.nlm.nih.gov/articles/PMC7859469/#Sec11>



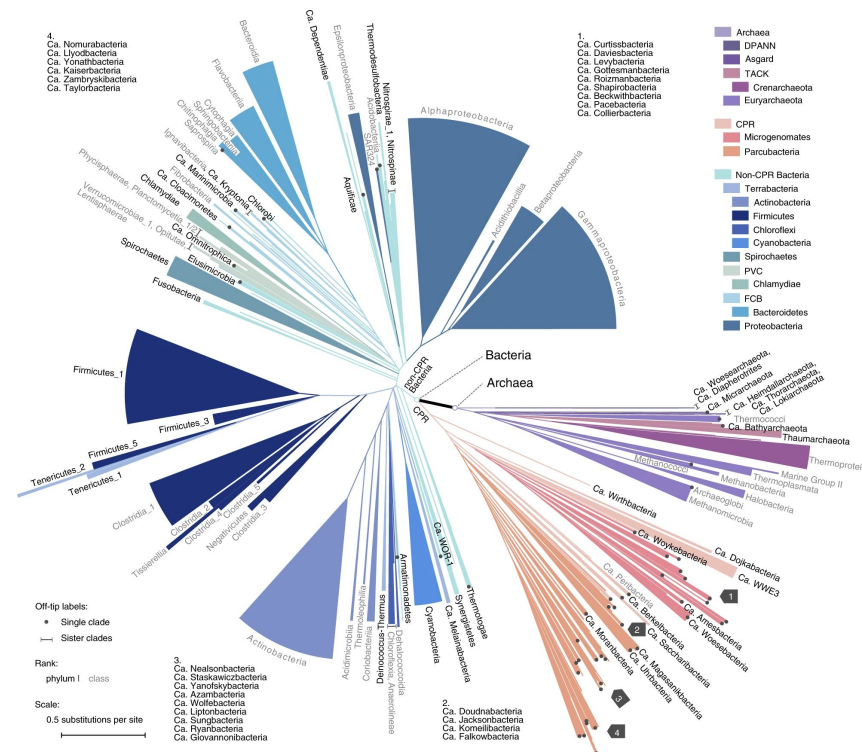
Gene tree topologies were reconstructed using the maximum likelihood (ML) method as implemented in the state-of-the-art phylogenetic inference program RAxML 8.2.10²⁶. The best amino acid substitution model for each of the 381 universal marker genes was inferred using RAxML's built-in script ProteinModelSelection.pl. Three phylogenetic trees were reconstructed for each gene family: one using a starting tree computed by the fast ML approach implemented in FastTree) and the other two using parsimony trees built with random seeds 12345 and 23456. RAxML was executed with the ML search convergence criterion (-D) and the CAT rate heterogeneity model without final optimization (-F) to reduce the execution time.

For each of the 1143 topologies (3×381), another RAxML run was executed to optimize branch lengths and to compute likelihood scores under the robust, but expensive Gamma rate heterogeneity model. Because of numerical instability, at least one of the RAxML runs failed for 39 of the 381 gene families. For those cases, IQ-TREE 1.6.1⁶¹, an alternative and faster maximum likelihood program, was used instead to optimize branch lengths using the same model (G4). The tree with the highest likelihood score among the three runs was retained for downstream applications. In 161 gene families, this tree was from the run with the FastTree starting tree, while in the remaining gene families the best tree was from either one of the random seeds.

<https://www.nature.com/articles/s41467-019-13443-4#Sec9>

[Qiyun Zhu](#), [Uyen Mai](#), [Wayne Pfeiffer](#), [Stefan Janssen](#), [Francesco Asnicar](#), [Jon G. Sanders](#), [Pedro Belda-Ferre](#), [Gabriel A. Al-Ghalith](#), [Evgenia Kopylova](#), [Daniel McDonald](#), [Tomasz Kosciolk](#), [John B. Yin](#), [Shi Huang](#), [Nimaichand Salam](#), [Jian-Yu Jiao](#), [Zijun Wu](#), [Zhenjiang Z. Xu](#), [Kalen Cantrell](#), [Yimeng Yang](#), [Erfan Sayyari](#), [Maryam Rabiee](#), [James T. Morton](#), [Sheila Podell](#), [Dan Knights](#), ... [Rob Knight](#) [+ Show authors](#)

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Ressources

conor.meehan@ntu.ac.uk <https://conmeehan.github.io/>
Phylogeny for the faint of heart: a tutorial (Sandra L. Baldauf)
<https://www.khanacademy.org/>
<https://nextstrain.org/>
<https://dunnlab.org/>

Thank you !

madeleine.violante@gmail.com

